💻 □ **Metropolitan University, Sylhet**

Department of Computer Science & Engineering

Lab Report – 04:

Coin Change - Minimum Number of Required Coins (Dynamic Programming)

Course Title: Algorithm Design and Analysis Lab

Course Code: CSE 232

Submitted To:

Honourable Sir, **Rishad Amin Pulok**
Senior Lecturer
Department of CSE
Metropolitan University, Sylhet

Submitted By:

Mahdi Hasan Shuvo
Student ID: 251-115-030
Batch: CSE 62nd (A)

Date of Submission:

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problem:

Coin Change - Minimum Number of Required Coins (Dynamic Programming)

CODE :

```
// Coin Change - Minimum Number of Required Coins (Dynamic Programming)
// Name: Mahdi Hasan Shuvo | ID: 251-115-030 | Batch: CSE 62nd (A)

#include<iostream>
#include<vector>
#include<algorithm>
#include<climits>
using namespace std;
int CoinChange(int n, vector <int> coins, int amount)
{
    vector <vector<int>> Table(n+1, vector <int> (amount+1, INT_MAX-1));

    for (int i = 0; i <= n; i++)
    {
        Table[i][0] = 0;
    }

    for (int i = 1; i <= n; i++)
    {
        for (int a = 1; a <= amount; a++)
        {
            if (coins[i-1] <= a)
            {
                Table[i][a] = min(Table[i-1][a], 1+Table[i][a-coins[i-1]]);
            }

            else
                Table[i][a] = Table[i-1][a];
        }
    }

    return Table[n][amount];
}



int main()
```

```cpp
{
    int n;
    cout << "Number of coins: ";
    cin >> n;

    vector <int> coins(n);

    for (int i = 0; i < n; i++)
    {
        cout << "Coin-" << i+1 << ": ";
        cin >> coins[i];
    }

    int amount;
    cout << "Amount: ";
    cin >> amount;

    int min_coins = CoinChange(n, coins, amount);

    cout << "Minimum required coins: " << min_coins << endl;

    return 0;
}
```

Output Output Output

OUTPUT:

```
1   // Coin Change - Minimum Number of Required Coins (Dynamic Programming)
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
Number of coins: 5
Coin-1: 4
Coin-2: 5
Coin-3: 6
Coin-4: 7
Coin-5: 8
Amount: 99
Minimum required coins: 13
PS G:\CSE-62\AD&A>
```

Thank You sir